

CONSTRUCTION OS

PRD - HR, Payroll & Workforce Module

Full Product Requirements Document for Product, UX/UI, Frontend, Backend, Data, QA, Security, DevOps, Finance and HR Operations Teams

Product purpose: Build the authoritative employment and workforce operating system for ConstructionOS - from recruitment and hiring through organisational assignment, attendance, labour deployment, payroll, competence and final offboarding.

Document control	Value
Document	PRD_HR_Payroll_Workforce_Module
Version	1.0
Status	Implementation-ready product specification
Audience	Full-stack development team and HR/payroll governance stakeholders
Primary routes	/hr, /hr/employees/:id, /hr/payroll/runs/:id, /hr/workforce
Access model	Permission-ready, tenant-isolated, company-scoped and field-confidential
Accounting policy	HR/Payroll calculates compensation; Finance owns journals and payment transactions
Workflow policy	Approval hooks only; central Approvals & Workflow Engine remains deferred
Integration policy	Self-contained initial release; government filing, bank and biometric integrations are future roadmap
Date	2026-08-05

Table of Contents

1. Executive Summary
2. Product Vision and Business Outcomes
3. Scope, Non-Goals and Assumptions
4. Domain Ownership and Source-of-Truth Boundaries
5. Personas, Roles and Access Model
6. Information Architecture
7. HR Dashboard
8. Employee Record and Employee Page
9. Employment Lifecycle
10. Organisation, Positions and Reporting Lines
11. Recruitment and Candidate Management
12. Offers and Pre-Employment
13. Onboarding
14. Employment Contracts
15. Compensation Components and History
16. Salary Bands, Grades and Job Families
17. Working Schedules
18. Shifts and Rosters
19. Attendance
20. Timesheets
21. Project Labour and Work Progress Integration
22. Leave and Entitlements
23. Absence and Sickness
24. Payroll Engine
25. Payroll Validation and Reconciliation
26. Payslips
27. Payroll-to-Finance Integration
28. Advances, Loans and Garnishments
29. Benefits
30. Expenses and Reimbursements
31. Training, Qualifications and Competence
32. Performance Management
33. Disciplinary and Grievance Cases
34. Workforce Planning and Capacity
35. External and Subcontractor Workforce
36. Asset Integration
37. Documents and Record Retention
38. Offboarding and Final Settlement
39. Intercompany Transfer and Multiple Employment
40. Calendar and Meetings Integration
41. Notifications and Workflow Readiness
42. Data Model
43. Backend and Event Architecture
44. API Contract
45. Frontend Architecture
46. Security, Privacy and Confidentiality
47. Employee Self-Service and Manager Workspace
48. Responsive and Offline UX
49. Premium Motion and Accessibility
50. Platform Configuration
51. Search, Filters and Saved Views
52. Reporting and Analytics
53. Validation, Errors and Concurrency
54. Non-Functional Requirements
55. Observability and Audit
56. QA Strategy and Test Matrix
57. Acceptance Criteria
58. Delivery Phases and Migration
59. Risks and Open Decisions
60. Page and Flow Registry

1. Executive Summary

The HR, Payroll & Workforce Module is the authoritative system for every worker's employment, compensation, attendance, availability, competence and organisational history across all ConstructionOS companies and projects. It combines HR master data, workforce operations and payroll calculation without duplicating User Profiles, Finance, Projects, Work Progress, HSE, Assets, Documents or Tasks.

The module is designed for multi-company construction groups that employ office staff, project teams, site crews, drivers, equipment operators, temporary workers, consultants and subcontractor personnel. It must support effective-dated employment records, project labour allocation, shift-based operations, multi-currency payroll, confidential field-level access and complete auditability.

Permanent separation: User Profile represents the person professionally. HR governs employment. Payroll calculates compensation. Work Progress records labour usage. HSE governs work eligibility. Finance records accounting and payment results.

1.1 Required business outcomes

- One controlled Employee Record for every employment relationship, with immutable history.
- No worker may be scheduled, paid or assigned outside valid employment, contract or competence dates.
- Every payroll value must be traceable to an effective rule and source record.
- Project labour cost must reconcile from timesheet to payroll result to Finance posting.
- Employees must receive safe self-service access without exposing restricted HR records.
- Offboarding must preserve historical work while revoking access and transferring active responsibility.
- Group reporting must consolidate companies without changing employer ownership.

1.2 Success indicators

Outcome	Target behavior
Employee integrity	No duplicate active employment relationship for the same person, company and period.
Payroll traceability	Every earnings or deduction line exposes formula, input, rate, quantity and source.
Workforce control	Managers can see capacity and eligibility without seeing restricted payroll or medical details.
Project costing	Actual labour hours and cost reconcile by employee, project, work package and cost code.
Privacy	Profile access never implies salary, payroll, medical or disciplinary access.
Operational resilience	The initial release operates without external payroll, bank or biometric integrations.

2. Product Vision and Business Outcomes

The product vision is to replace disconnected employee spreadsheets, paper attendance, informal shift rosters, manual payroll calculations and isolated project labour logs with a governed enterprise workforce platform. The system must remain practical for field operations while meeting payroll, privacy, audit and multi-company requirements.

2.1 Product principles

1. Effective-dated records: employment, position, compensation, schedules and payroll rules change through dated records rather than overwrites.
2. Source ownership: linked modules remain authoritative for their domains.
3. Calculation transparency: no unexplained payroll total is acceptable.
4. Privacy by design: sensitive fields and documents use independent access controls.

5. Operational eligibility: workers are assignable only when employment, schedule, competence and HSE gates are satisfied.
6. Configuration over hard-coding: rules vary by company, location and jurisdiction.
7. Immutable payroll: locked results are corrected through adjustments, reversals or reopened periods with audit history.
8. No automated performance judgment from attendance, tasks or activity data.

2.2 Core questions

- Who employs the worker and under what contract, position, grade and schedule?
- Where and when did the worker actually work?
- What should be paid for the payroll period and why?
- Which project, department or cost centre consumed the labour?
- Is the worker qualified, inducted and available for the planned assignment?
- What leave, overtime, benefits, loans or deductions apply?
- What must happen when the worker joins, transfers, changes compensation or leaves?

3. Scope, Non-Goals and Assumptions

3.1 In scope

- Employee and employment records, organisation, positions, reporting lines and history.
- Recruitment, candidate management, offers, onboarding and pre-employment checks.
- Employment contracts, compensation, salary bands, benefits and allowances.
- Working schedules, shifts, rosters, attendance, timesheets and project labour allocation.
- Leave, absence, sickness and operational fitness status.
- Payroll groups, rules, calculation, validation, payslips, final settlement and Finance export.
- Training, qualifications, competence, performance reviews, disciplinary and grievance cases.
- Workforce planning, external workforce, offboarding and intercompany transfers.
- Data model, APIs, events, responsive UX, offline drafts, security, audit and testing.

3.2 Out of scope for v1

- Jurisdiction-specific statutory filing integrations.
- Bank payment-file transmission or live banking.
- Biometric devices and mandatory continuous GPS tracking.
- Clinical medical record management.
- Central approval orchestration, which remains deferred to the final Approvals & Workflow PRD.
- Automatic employee performance ratings from platform activity.
- AI-generated hiring, termination or disciplinary decisions.
- External job-board, learning-management or government payroll integrations.

3.3 Assumptions

Assumption	Implication
Multi-tenant and multi-company	Every employment, payroll and document record carries tenant and legal-company scope.
User Profiles exist	Employee records link to professional profiles without exposing confidential HR fields.
Documents exists	Contracts, certificates, payslips and evidence are controlled through Document IDs.
Finance exists	Payroll exports posting-ready results; Finance remains the ledger and payment source of truth.
HSE and Work Progress exist	Eligibility and labour-usage integration contracts can be implemented.
Approvals deferred	Decision states and hooks are implemented without a universal workflow designer.

4. Domain Ownership and Source-of-Truth Boundaries

Domain	Source of truth	HR/Payroll responsibility
Authentication	User Account	Link employee to account; never store passwords or MFA data.
Professional identity	User Profile	Publish permitted employment and competence projections.
Employment	HR	Own employer, contract interpretation, position, status and history.
Collaboration teams	Teams	Read team membership; never change employer or department through team assignment.
Project records	Projects	Reference project assignments and labour demand.
Physical labour usage	Work Progress	Receive actual labour context; HR owns payable time and cost rate.
Operational eligibility	HSE	Publish competence status and receive work restrictions.
Equipment assignments	Assets	Display linked asset projections and offboarding return status.
Files and revisions	Documents	Link controlled contracts, payslips and certificates.
Accounting and payment	Finance	Export payroll journals and receive posting/payment state.
Execution work	Tasks	Create onboarding, corrective and offboarding tasks.
Approvals	Future Approvals Engine	Store decision hooks and references only.

Payroll boundary: HR/Payroll calculates gross-to-net and labour allocation. Finance owns posted journals, bank transactions, tax payments, supplier payments and financial statements.

5. Personas, Roles and Access Model

5.1 Primary personas

Persona	Primary needs
Group HR Director	Cross-company headcount, workforce planning, policy and consolidated reporting.
Company HR Manager	Employee lifecycle, contracts, leave, cases and local compliance.
Payroll Manager	Payroll inputs, calculation, validation, locking and Finance export.
Recruiter	Vacancies, candidates, interviews, offers and conversion.
Workforce Planner	Demand, capacity, rosters, skills and project allocation.
Line Manager	Team availability, timesheets, leave, reviews and onboarding progress.
Project/Site Manager	Eligible labour, shifts, actual hours and project allocation.

Persona	Primary needs
HSE Manager	Training, certification and operational eligibility projection.
Employee	Self-service access to own permitted records and submissions.
Finance User	Payroll summaries, posting state and reimbursement references.
Auditor	Scoped read-only access to history and calculation evidence.

5.2 Access dimensions

- Tenant, corporate group, legal company and branch.
- Organisation unit, position, manager relationship and project.
- Employee self-access and explicit HR/payroll role.
- Record domain and confidentiality zone.
- Field-level access for salary, bank, identification, medical, disciplinary and executive compensation.
- Effective-date and historical access where required.

5.3 Confidentiality zones

Zone	Examples
Public professional	Name, photo, job title and public skills.
Internal professional	Department, manager, project assignments and work contact.
HR confidential	Employment contract, address, leave and personal records.
Payroll restricted	Salary, bank, payslip, deductions and payroll results.
Medical restricted	Certificates and operational fitness details with minimum disclosure.
Disciplinary restricted	Investigations, warnings and grievance evidence.
Executive compensation	Board/executive compensation and benefits.
Legal hold	Records preserved under a legal matter.

6. Information Architecture

6.1 Main navigation

- HR Dashboard
- Employees
- Organisation
- Positions
- Workforce Planning
- Recruitment
- Candidates
- Vacancies
- Onboarding
- Employment Contracts
- Attendance
- Shifts & Rosters
- Timesheets
- Leave & Absence
- Payroll
- Compensation & Benefits
- Expenses & Reimbursements
- Training & Competence
- Performance
- Disciplinary & Grievances

- Offboarding
- External Workforce
- HR Documents
- Reports
- Settings

6.2 Route hierarchy

```

/hr
  /dashboard
  /employees/:employeeId
  /organisation
  /positions/:positionId
  /recruitment/vacancies/:vacancyId
  /candidates/:candidateId
  /onboarding/:planId
  /attendance
  /rosters
  /timesheets/:timesheetId
  /leave/requests/:requestId
  /payroll/runs/:runId
  /workforce
  /training
  /cases/:caseId
  /offboarding/:planId
  /settings

```

6.3 Global behaviors

- All records use permanent business identifiers and complete activity timelines.
- Lists support global search, filters, saved views, column chooser, export and safe bulk operations.
- Detail pages display effective-date history and linked source records.
- Confidential sections are hidden or masked independently from page-level access.
- Archived records remain addressable from historical links and audit timelines.

7. HR Dashboard

7.1 KPI groups

KPI group	Required KPIs
Headcount	Active employees by company, department, project, status and employment type.
Workforce today	Present, on leave, missing attendance, overtime exposure and unavailable.
Employment	New starters, probation ending, contracts expiring and offboarding.
Recruitment	Open positions, time to fill, candidates by stage and accepted offers.
Payroll	Run status, validation errors, gross payroll, employer cost and variance.
Competence	Training compliance, certification expiry and blocked eligibility.
Project labour	Actual hours, project labour cost, allocation gaps and overcapacity.

7.2 Functional requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-DB-001	Dashboard	Every KPI must drill into a pre-filtered source list.	Must	Clicked KPI opens matching source records.
HR-DB-002	Dashboard	Critical payroll and eligibility warnings must appear above trend analytics.	Must	Critical items are visible without scrolling on desktop.
HR-DB-003	Dashboard	Group consolidation must preserve company ownership and currency context.	Must	Totals reconcile to scoped company source values.
HR-DB-004	Dashboard	Salary and payroll KPIs require independent permission checks.	Must	Unauthorized users see no inferred values.
HR-DB-005	Dashboard	Users may save private dashboard filters and time ranges.	Should	Preferences restore per user.

8. Employee Record and Employee Page

8.1 Employee identity and employment fields

Field group	Required fields
Identity	Permanent employee ID, employee number, linked person/profile, legal/preferred name, photo.
Personal	Date of birth, contact, address, nationality, identity and work authorization references.
Employment	Employer, branch, department, position, grade, manager, status, type, start/end dates.
Schedule	Work location, cost centre, standard schedule and time zone.
Confidential	Emergency contacts, bank/tax references, medical fitness projection and access classification.

8.2 Employee Page sections

9. Overview
10. Employment
11. Contracts
12. Position and department history
13. Compensation
14. Attendance
15. Timesheets
16. Leave
17. Projects and labour allocation
18. Training and qualifications
19. Performance
20. Disciplinary and grievances
21. Assigned assets
22. Expenses
23. Documents
24. Onboarding and offboarding
25. Emergency details
26. Activity timeline

8.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-EMP-001	Employee	Employee number must be unique within configured company scope.	Must	Duplicate create is rejected.
HR-EMP-002	Employee	An Employee Record may link to one User Profile without exposing confidential fields.	Must	Profile API returns only permitted projections.
HR-EMP-003	Employee	Employment changes create effective-dated history rather than overwriting prior values.	Must	Historical date query returns previous state.
HR-EMP-004	Employee	Sensitive identity and bank values must support masking.	Must	Unauthorized views show masked or no value.
HR-EMP-005	Employee	Employee archive must preserve payroll, project and audit links.	Must	Historical records remain resolvable.

9. Employment Lifecycle

Candidate -> Offer -> Pre-employment -> Hired -> Onboarding -> Active -> Probation Review -> Confirmed -> Transfer/Promotion -> Notice Period -> Offboarding -> Terminated -> Former Employee -> Archived
 Alternative: Suspended | Long-Term Leave | Secondment | Retired | Deceased | No Show | Offer Withdrawn | Offer Declined

9.1 Lifecycle controls

- Only configured transitions are valid.
- Every transition stores actor, timestamp, effective date, reason and source document.
- Backdated changes trigger impact analysis for payroll, attendance, leave, project allocation and access.
- Termination cannot delete historical records or assignments.
- Rehire creates a new Employment Relationship while preserving the previous one.

9.2 Employment status requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-LC-001	Lifecycle	Active status requires a valid employer, position or allowed position exception, start date and contract state.	Must	Invalid activation returns structured validation errors.
HR-LC-002	Lifecycle	Termination date must stop future roster, attendance and payroll eligibility unless an authorized exception exists.	Must	Future assignments are blocked or flagged.
HR-LC-003	Lifecycle	Rehire must create a new relationship rather than reopen old payroll history.	Must	Previous employment remains closed and queryable.

10. Organisation, Positions and Reporting Lines

10.1 Organisation hierarchy

Corporate Group -> Legal Company -> Branch -> Division -> Department -> Section -> Position -> Employee

10.2 Position model

Field group	Fields
Identity	Position code, title, company, department and reporting position.
Work design	Grade, job family, employment category, location and cost centre.
Capacity	Planned headcount, occupied count, vacancy state and allocation.
Requirements	Skills, qualifications, certifications, HSE and project eligibility.
Compensation	Salary band, overtime, allowance and bonus eligibility.
Lifecycle	Planned, vacant, occupied, temporarily covered, frozen or abolished.

10.3 Organisation requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-ORG-001	Organisation	An employee may have one primary position per employment relationship and effective period.	Must	Overlapping primary assignments are rejected.
HR-ORG-002	Organisation	Acting and secondary assignments must be explicitly marked and dated.	Must	They do not replace primary assignment history.
HR-ORG-003	Organisation	Reporting cycles in the position hierarchy must be prevented.	Must	Cyclic assignment returns validation error.
HR-ORG-004	Organisation	Headcount is derived from position capacity and active assignments.	Should	Dashboard values reconcile to positions.

11. Recruitment and Candidate Management

11.1 Vacancy Page

- Vacancy code, position, company, department, hiring manager and recruiter.
- Employment type, location, headcount, salary range, skills and certificates.
- Opening date, target hire date, project requirement, source and status.
- Job description, interview plan, assessments and candidate pipeline.

11.2 Candidate Page

- Identity, contact, CV, portfolio, qualifications, skills, languages and work history.
- Applications, interviews, assessments, references, expected salary and availability.
- Offer history, duplicate detection, consent/retention status, documents and timeline.

11.3 Recruitment pipeline

Applied -> Screening -> Shortlisted -> Interview -> Assessment -> Reference Check -> Offer Preparation
 -> Offer Sent -> Accepted -> Hired
 Alternative: Rejected | Withdrawn | Talent Pool | Offer Declined | On Hold | Duplicate

11.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-REC-001	Recruitment	Candidate duplicates must be detected using configurable identity and contact rules.	Must	Potential matches require user review before creation.
HR-REC-002	Recruitment	Candidate conversion preserves recruitment history and links documents without duplication.	Must	Converted employee links back to candidate.
HR-REC-003	Recruitment	Candidate retention and consent state must be configurable.	Must	Expired records follow configured retention action.
HR-REC-004	Recruitment	Hiring decisions must never be generated automatically from protected or sensitive data.	Must	No prohibited automated decision endpoint exists.

12. Offers and Pre-Employment

12.1 Offer model

Category	Fields
Employment	Position, employer, start date, type, schedule and work location.
Compensation	Base salary, currency, frequency, allowance, benefits and bonus conditions.
Terms	Probation, notice, working hours, offer expiry and conditions precedent.
Governance	Template, signatory, candidate decision, revisions and documents.

12.2 Pre-employment checks

- Identity and right-to-work verification.
- Qualification, licence and reference verification.
- Background-check status where permitted.
- Operational fitness status, bank/tax details and emergency contact.
- Signed policies and required documents.

12.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-OFR-001	Offers	Accepted offer values must be preserved as a versioned snapshot.	Must	Later edits create a new version.

ID	Area	Requirement	Priority	Acceptance signal
HR-OFR-002	Pre-employment	Missing mandatory conditions must block Hire transition.	Must	UI lists unresolved conditions.
HR-OFR-003	Pre-employment	Restricted medical details must not be visible to recruiters or general managers.	Must	Only operational fitness projection is exposed.

13. Onboarding

13.1 Onboarding plan

- Employee, position, start date, company, department, manager, HR owner and project/site.
- Template, checklist items, dependencies, completion percentage and readiness state.
- Links to Tasks, User Account, Assets, HSE, Documents, Teams and Projects.

13.2 Checklist categories

- Employment and payroll documents
- User account and system access
- Equipment and workplace setup
- Policies and data protection
- HSE/site induction
- Role training
- Manager and team introduction
- Project assignment
- Probation objectives

13.3 Source ownership

Item	Owner
Onboarding requirement and readiness	HR
Executable checklist work	Tasks
Account activation and authentication	User Account
Equipment assignment	Assets
Induction and work eligibility	HSE
Signed files and acknowledgements	Documents
Project/team membership	Projects / Teams

13.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-ONB-001	Onboarding	Onboarding templates are versioned and company/position configurable.	Must	Plan records template version.
HR-ONB-002	Onboarding	Checklist completion must not silently grant platform permissions.	Must	Access remains controlled by authorization services.

ID	Area	Requirement	Priority	Acceptance signal
HR-ONB-003	Onboarding	Readiness shows incomplete blocking and non-blocking items separately.	Must	Header explains blockers.

14. Employment Contracts

14.1 Contract fields

- Contract number, employer, employee, position and department.
- Start/end, probation, schedule, work location, salary and pay frequency.
- Overtime terms, leave entitlement, benefits, notice and governing law.
- Signed status, amendments, renewal date and linked controlled documents.

14.2 Lifecycle

Draft -> Review Ready -> Signature Pending -> Signed -> Active -> Renewal Due -> Renewed -> Expired -> Terminated -> Closed

14.3 Contract boundaries

The Contracts Module may govern the formal legal agreement lifecycle. HR owns the operational employment interpretation, current employment state and payroll effect. Signed files are immutable and changes require amendments.

14.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-CON-001	Contracts	Active employment must reference a valid contract or configured exception.	Must	Activation fails without required reference.
HR-CON-002	Contracts	Signed contract files and key term snapshots are immutable.	Must	Changes require amendment version.
HR-CON-003	Contracts	Renewal and expiry create Calendar projections and notification-ready events.	Must	Source-linked events are emitted.

15. Compensation Components and History

15.1 Supported components

Type	Examples
Earnings	Base salary, hourly/daily wage, overtime, shift premiums, project/site/travel/meal/housing/hazard allowances, bonus, commission, back pay.
Deductions	Tax, social contribution, pension, health insurance, advance, loan, absence deduction, garnishment and custom deductions.
Employer cost	Employer tax, employer contributions, insurance, pension, benefits and custom employer costs.

15.2 Effective-dated compensation

- Previous and new value
- Effective date and end date
- Currency and pay frequency
- Reason and source document
- Changed by and approval reference
- Tax, overtime and allowance treatment

15.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-CMP-001	Compensation	Overlapping active base-pay records for one employment relationship are prohibited.	Must	Save rejects overlap.
HR-CMP-002	Compensation	Historical payroll uses the rule effective in the payroll period.	Must	Recalculation reproduces prior result.
HR-CMP-003	Compensation	Compensation changes after payroll lock trigger retroactive adjustment, not silent rewrite.	Must	Adjustment references original result.
HR-CMP-004	Compensation	All monetary fields carry currency and rounding policy.	Must	No currency-less amount is persisted.

16. Salary Bands, Grades and Job Families

- Grade and job family
- Company and location
- Currency
- Minimum, midpoint and maximum
- Effective dates
- Position mappings
- Overtime, allowance and bonus eligibility

Boundary: Salary bands guide compensation and control warnings. They do not post payroll transactions and do not automatically change employee pay.

ID	Area	Requirement	Priority	Acceptance signal
HR-BND-001	Salary bands	Bands are effective-dated and may vary by company, location and currency.	Must	Historical queries return correct band.
HR-BND-002	Salary bands	Out-of-band compensation generates a warning, not an automatic correction.	Should	Authorized user can proceed with reason.

17. Working Schedules

17.1 Schedule types

- Standard office
- Fixed shift
- Rotating shift
- Flexible
- Part-time
- Night
- Project/site schedule
- Seasonal
- Remote/hybrid
- Custom

17.2 Versioned schedule fields

- Working days and start/end times
- Break rules and paid/unpaid treatment
- Weekly hours and daily maximum
- Overtime threshold
- Night period and weekend rules
- Holiday calendar
- Time zone
- Effective dates

17.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-SCH-001	Schedules	Schedules are versioned and time-zone aware.	Must	DST and effective-date tests pass.
HR-SCH-002	Schedules	Cross-midnight work must be allocated correctly to attendance and payroll dates.	Must	Configured policy produces deterministic result.
HR-SCH-003	Schedules	Schedule changes recalculate future rosters only unless explicitly applied retroactively.	Must	Historical attendance is preserved.

18. Shifts and Rosters

18.1 Shift definition

- Shift code/name
- Start/end and cross-midnight rule
- Breaks and paid hours
- Shift premium
- Required role and competence
- Site/project and capacity
- Setup and handover buffers

18.2 Roster views

- Employee timeline
- Team timeline
- Project/site timeline
- Unfilled positions
- Conflicts and leave overlays
- Competence warnings
- Maximum-hours and overtime projection

18.3 Conflict rules

- Overlapping shift
- Insufficient rest
- Approved leave
- Expired competence
- Employment outside dates
- Site/HSE restriction
- Maximum legal/configured hours
- Allocation elsewhere

18.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-RST-001	Rosters	Hard conflicts block assignment unless an authorized override policy exists.	Must	Conflict response identifies rule and source.
HR-RST-002	Rosters	Roster changes after publication generate a versioned change and notification event.	Must	Previous roster remains auditable.
HR-RST-003	Rosters	Bulk assignment must be transactional per configured batch policy.	Must	Partial failure is reported and retry-safe.

19. Attendance

19.1 Sources and events

- Manual entry
- Web/mobile check-in
- Optional GPS or QR/site check-in
- Imported attendance
- Shift confirmation
- Approved correction
- Check-in/out, break start/end, site transfer and missed events

19.2 Daily states

- Present
- Absent
- Partial day
- Late
- Left early
- On leave
- Sick
- Training
- Business travel
- Remote
- Rest day
- Public holiday
- Suspended
- Unknown

19.3 Anomaly rules

- Missing check-out
- Duplicate check-in
- Impossible travel

- Outside assigned location
- Excessive duration
- Attendance during leave
- Attendance after termination
- Attendance without shift
- Clock/device anomaly

19.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-ATT-001	Attendance	Every raw attendance event is immutable and corrections create adjustment records.	Must	Original event remains queryable.
HR-ATT-002	Attendance	GPS is optional, purpose-limited and configurable by company/site.	Must	No location collection occurs when disabled.
HR-ATT-003	Attendance	Offline events preserve device and server timestamps and duplicate-detection keys.	Must	Retries do not create duplicate attendance.
HR-ATT-004	Attendance	Daily state is derived reproducibly from events, schedule, leave and policy.	Must	Rebuild produces same state.

20. Timesheets

20.1 Timesheet levels

- Employee daily/weekly
- Project timesheet
- Work-package timesheet
- Team timesheet
- Equipment-operator timesheet
- Payroll-period timesheet

20.2 Timesheet line

- Employee, date, start/end, regular/overtime and break
- Company, department, project, structure/zone, work package, cost code and activity
- Shift, source, notes and status

20.3 Lifecycle

Draft -> Submitted -> Reviewed -> Verified -> Payroll Locked -> Exported -> Archived
 Alternative: Returned | Rejected | Corrected | Superseded

20.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-TMS-001	Timesheets	Lines cannot allocate hours outside active employment without authorized exception.	Must	Invalid lines are rejected.

ID	Area	Requirement	Priority	Acceptance signal
HR-TMS-002	Timesheets	Payroll-locked lines are immutable; corrections use adjustment lines.	Must	Original and adjustment reconcile.
HR-TMS-003	Timesheets	Attendance and timesheet differences are visible but do not silently overwrite either source.	Must	Variance view shows both values.
HR-TMS-004	Timesheets	One hour cannot be double allocated to overlapping projects unless policy permits split allocation.	Must	Overlap validation applies.

21. Project Labour and Work Progress Integration

21.1 Data exchange

Work Progress provides	HR provides
Physical activity and work package	Employee and employer identity
Location, project, zone and shift	Position, role and eligibility
Crew and actual labour hours	Schedule, payable classification and cost rate
Completed quantities and evidence	Payroll status and labour cost allocation

21.2 Permanent rule

Source of truth: Work Progress records where labour was used. HR records employment and payable time. Payroll calculates compensation. Finance records the financial cost.

21.3 Labour allocation fields

- Employee
- Project/work package
- Role
- Allocation percentage
- Planned/actual hours
- Start/end
- Cost rate and billable rate
- Overtime
- Source timesheet
- Calculated labour cost

21.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-LAB-001	Project labour	Labour allocations must reconcile to timesheet hours by configured tolerance.	Must	Variance report identifies mismatch.
HR-LAB-002	Project labour	Cost rates are effective-dated and access-restricted.	Must	Unauthorized project users see no salary inference.

ID	Area	Requirement	Priority	Acceptance signal
HR-LAB-003	Project labour	Employee can work across projects while payroll remains employer-owned.	Must	Allocation does not change company employment.

22. Leave and Entitlements

22.1 Leave types and balances

- Annual, sick, unpaid, maternity, paternity, parental, bereavement, study, marriage, medical, public duty, compensatory, time-off-in-lieu, work injury and custom.
- Opening balance, accrued, used, planned, adjusted, expired, carried forward and available.

22.2 Request lifecycle

Draft -> Submitted -> Decision Pending -> Approved -> Scheduled -> Taken -> Closed
 Alternative: Returned | Rejected | Cancelled | Partially Cancelled

22.3 Validation

- Balance and employment dates
- Overlap
- Holiday calendar and schedule
- Minimum notice
- Consecutive day limit
- Restricted project period
- Team capacity warning
- Required attachment

22.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-LEV-001	Leave	Accrual rules are versioned and reproducible.	Must	Balance rebuild matches stored ledger.
HR-LEV-002	Leave	Leave balance uses an immutable transaction ledger, not only a mutable total.	Must	Every balance change has a source transaction.
HR-LEV-003	Leave	Calendar/availability projection hides the leave reason unless permitted.	Must	Managers see availability only.
HR-LEV-004	Leave	Cancelled or corrected leave creates reversal transactions.	Must	Balance audit remains complete.

23. Absence and Sickness

23.1 Absence case

- Employee, type, start/end and partial-day detail
- Notification and expected/actual return
- Certificate requirement

- Work-related indicator
- Restricted medical category
- Pay treatment
- Return-to-work review
- Documents and status

23.2 Privacy model

Operational users may see	Restricted HR/medical users may see
Availability state	Diagnosis or clinical detail where lawfully recorded
Expected return date	Medical documents
Operational work restrictions	Treatment information
Certificate received/missing	Full confidential notes

23.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-ABS-001	Absence	Operational fitness status must be separately stored from medical detail.	Must	HSE/manager APIs return minimum data.
HR-ABS-002	Absence	Absence pay treatment must be traceable to configured rules.	Must	Payroll line references rule and case.
HR-ABS-003	Absence	Return-to-work review may create Tasks without exposing medical attachments.	Must	Task contains only permitted operational content.

24. Payroll Engine

24.1 Payroll structures

- Payroll group
- Pay calendar and frequency
- Payroll period
- Company and jurisdiction configuration
- Currency
- Employee population
- Calculation rules
- Payment date

24.2 Payroll run lifecycle

Period Open -> Input Collection -> Preliminary Calculation -> Validation -> Corrections -> Final Calculation -> Payroll Locked -> Payslips Generated -> Finance Posted -> Payment Prepared -> Paid -> Period Closed
 Alternative: Failed | Reopened | Reversed | Partially Paid | Cancelled

24.3 Inputs

- Effective compensation
- Attendance and timesheets
- Overtime and shift premiums
- Leave and absence

- Allowances, bonuses and commissions
- Expenses, advances, loans and garnishments
- Benefits, tax and social contributions
- Manual/retroactive adjustments
- Termination settlement

24.4 Calculation order

27. Resolve active employment and payroll group.
28. Resolve effective compensation and schedule.
29. Resolve payable time.
30. Calculate regular earnings.
31. Calculate premiums and overtime.
32. Apply taxable benefits.
33. Calculate employee deductions.
34. Calculate taxes and contributions.
35. Calculate employer costs.
36. Apply advances, loans and garnishments.
37. Calculate net pay.
38. Allocate labour cost.
39. Validate and persist immutable result.

24.5 Calculation trace

- Rule/version
- Source record
- Input
- Rate
- Quantity
- Formula
- Result
- Currency
- Rounding
- Adjustment history

24.6 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-PAY-001	Payroll	Calculation must be deterministic for the same versioned inputs and rules.	Must	Repeated run produces identical result.
HR-PAY-002	Payroll	Every result line must expose a human-readable calculation trace.	Must	UI can explain each value.
HR-PAY-003	Payroll	Locked results are immutable.	Must	Direct update is rejected.
HR-PAY-004	Payroll	Retroactive changes create adjustment lines linked to the original result.	Must	Audit shows before/after and affected period.
HR-PAY-005	Payroll	Payroll runs are isolated by legal company and currency policy.	Must	Cross-company contamination tests pass.
HR-PAY-006	Payroll	Calculation jobs must be idempotent and resumable.	Must	Retry does not duplicate results.

25. Payroll Validation and Reconciliation

25.1 Critical validations

- Active employment and company
- Valid compensation and payroll group
- No duplicate result
- Verified/locked time input
- No double-counted overtime
- Valid leave/absence treatment
- Gross-to-net reconciliation
- Tax/contribution reconciliation
- Employer-cost reconciliation
- Currency and bank completeness
- Negative net handling
- Large variance
- Termination date compliance

25.2 Variance controls

- Previous period
- Employee historical average
- Contracted salary
- Expected hours
- Department budget
- Project labour budget

25.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-VAL-001	Validation	Hard validation errors block payroll lock; warnings require recorded resolution.	Must	Lock endpoint lists unresolved items.
HR-VAL-002	Validation	Validation rules are versioned with effective dates.	Must	Historical run retains rule version.
HR-VAL-003	Validation	Variance warnings must not imply fraud or employee wrongdoing.	Must	Labels remain factual and source-based.
HR-VAL-004	Reconciliation	Payroll totals must reconcile by employee, component, company, project and Finance export.	Must	Reconciliation report has zero unexplained variance.

26. Payslips

26.1 Required sections

- Employer and employee
- Payroll period and pay date
- Earnings, overtime, allowances and benefits
- Deductions, taxes and contributions
- Net pay
- Employer costs where configured
- Leave balance

- Year-to-date values
- Bank destination
- Notes

26.2 Document requirements

- Controlled template and language
- Stored through Documents
- Employee self-service visibility
- Issued revision preserved
- Correction/replacement without deleting history
- Confidential watermark and access policy

ID	Area	Requirement	Priority	Acceptance signal
HR-PSL-001	Payslips	A payslip may be generated only from a locked payroll result.	Must	Draft runs cannot issue payslips.
HR-PSL-002	Payslips	Replacement payslip references and supersedes the original.	Must	Both revisions remain accessible to authorized users.
HR-PSL-003	Payslips	Employees may access only their own payslips unless explicitly authorized otherwise.	Must	Horizontal access tests pass.

27. Payroll-to-Finance Integration

27.1 Export payload

- Payroll payable
- Net salary payable
- Tax and social contribution payable
- Benefit payable
- Employer expense
- Department/project/cost-centre allocation
- Currency
- Payroll run reference
- Reversal/adjustment references

27.2 Finance response

- Journal ID and posting status
- Payment status
- Bank transaction
- Reversal reference
- Rejected line details

27.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-FIN-001	Finance integration	Payroll export must be balanced and idempotent.	Must	Duplicate submission returns same Finance reference.
HR-FIN-002	Finance integration	Finance rejection must not unlock payroll automatically.	Must	User receives structured error and remediation path.
HR-FIN-003	Finance integration	Reopened payroll after posting requires reversal or adjustment linkage.	Must	No orphaned journals remain.

ID	Area	Requirement	Priority	Acceptance signal
HR-FIN-004	Finance integration	Finance receives summaries and dimensions, not unrelated confidential HR data.	Must	Payload schema contains only approved fields.

28. Advances, Loans and Garnishments

Record	Required data
Advance	Requested/approved amount, issue date, repayment method, deductions, balance and Finance payment.
Loan	Principal, interest, term, installments, dates, balance, payroll and early repayment.
Garnishment	Authority, legal document, priority, amount/percentage, protected minimum, dates and confidentiality.

ID	Area	Requirement	Priority	Acceptance signal
HR-LON-001	Loans	Repayment schedules must reconcile to deductions and outstanding balance.	Must	Ledger rebuild matches balance.
HR-LON-002	Garnishments	Priority and protected-minimum rules are configurable and versioned.	Must	Payroll trace shows applied rule.
HR-LON-003	Advances	Finance owns disbursement; Payroll owns repayment treatment.	Must	Links exist in both directions.

29. Benefits

- Health, pension and life insurance
- Company car and housing
- Meal, phone, travel and education
- Childcare and wellness
- Equity reference
- Custom benefit

29.1 Benefit record

- Employee
- Provider and plan
- Coverage and dependants
- Start/end
- Employee/employer contribution
- Tax treatment
- Finance reference
- Documents

ID	Area	Requirement	Priority	Acceptance signal
HR-BEN-001	Benefits	Benefits are effective-dated and payroll treatment is versioned.	Must	Historical payroll uses correct benefit.
HR-BEN-002	Benefits	Dependants and sensitive coverage fields require separate access.	Must	General managers cannot read restricted detail.

30. Expenses and Reimbursements

30.1 Expense fields

- Employee
- Date/category
- Project/cost centre
- Currency/amount/tax
- Receipt
- Business reason
- Mileage
- Advance offset
- Policy exception
- Status
- Finance reimbursement reference

30.2 Ownership

HR supplies employee eligibility and employment context. Finance owns payment and accounting. Formal approval chains remain deferred to the Approvals Engine.

ID	Area	Requirement	Priority	Acceptance signal
HR-EXP-001	Expenses	Receipt and policy evidence are stored through Documents.	Must	Expense record links Document ID.
HR-EXP-002	Expenses	Expense status must reconcile to Finance reimbursement state.	Must	UI distinguishes submitted, approved-context and paid.
HR-EXP-003	Expenses	Duplicate receipt detection creates a warning and review flow.	Should	Potential duplicates are linked.

31. Training, Qualifications and Competence

31.1 Training catalogue and records

- Course, provider, category, delivery method, duration and validity
- Required roles/projects, assessment, certificate, cost and renewal
- Employee schedule, completion, result, score, trainer and expiry

31.2 Competence matrix

- Position requirement
- Required skill/level
- Required certificate
- Employee level and verification
- Gap

- Training plan
- Work restriction

31.3 HSE projection

- Eligible
- Eligible with restrictions
- Training missing
- Certificate expired
- Not eligible

31.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-TRN-001	Training	Certificate expiry must generate Calendar and notification-ready events.	Must	Events contain source link.
HR-TRN-002	Competence	Eligibility is derived from versioned requirements and verified records.	Must	Result explains missing requirement.
HR-TRN-003	Competence	HSE receives only eligibility and necessary qualification detail.	Must	No unrelated salary/medical data in API.
HR-TRN-004	Training	Expired competence may block roster or high-risk assignment according to configuration.	Must	Conflict is enforced consistently.

32. Performance Management

32.1 Review types and structure

- Probation, annual, project, role-change, development, improvement and custom reviews.
- Period, reviewer, objectives, competencies, achievements, development needs, comments, rating, future objectives and training plan.

32.2 Safeguards

No automated judgment: Task completion, attendance, Teams KPIs, productivity or platform activity must not automatically become a performance rating or disciplinary conclusion.

ID	Area	Requirement	Priority	Acceptance signal
HR-PER-001	Performance	Review templates and rating scales are versioned.	Must	Review stores template version.
HR-PER-002	Performance	Employee and manager comments retain authorship and finalization history.	Must	Finalized review is immutable except addendum.
HR-PER-003	Performance	Source metrics displayed in reviews must remain factual and drillable.	Must	No hidden algorithmic score.

33. Disciplinary and Grievance Cases

33.1 Disciplinary case

- Case code, employee, allegation, date and policy
- Investigator, evidence, meetings and employee response
- Findings, outcome, warning, effective period, appeal and confidentiality

33.2 Grievance case

- Employee, complaint, parties, confidentiality and investigation
- Meetings, findings, resolution, appeal and retaliation monitoring

33.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-CAS-001	Cases	Case access is independent from general Employee Page access.	Must	Unauthorized employee/manager access is denied.
HR-CAS-002	Cases	Evidence must preserve uploader, timestamp, source and document hash/reference.	Must	Audit is complete.
HR-CAS-003	Cases	Outcomes and warnings are effective-dated and cannot be silently deleted.	Must	Superseded state remains visible.
HR-CAS-004	Cases	Case data must not automatically influence performance scoring.	Must	No implicit scoring integration.

34. Workforce Planning and Capacity

34.1 Demand and supply

Demand planning	Supply planning
Company, department, project and work package	Available employees and current allocation
Role, skill, headcount and required hours	Leave, training and schedule availability
Dates, shift, location and priority	Competence, access, cost and employment limits

34.2 Capacity states

- Available
- Partially available
- Fully allocated
- Overallocated
- On leave
- Training
- Restricted
- Unavailable
- Unknown

34.3 Formula

Available capacity = Contracted working capacity - Confirmed allocations - Approved leave - Training -

Configured non-project time

34.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-WFP-001	Planning	Capacity calculations expose every source value and effective date.	Must	UI explains result.
HR-WFP-002	Planning	Over-allocation creates warning and does not silently change assignments.	Must	Manager resolves explicitly.
HR-WFP-003	Planning	Cost visibility is permission-controlled and may use standard cost rate projection.	Must	Salary cannot be inferred by unauthorized user.

35. External and Subcontractor Workforce

35.1 Worker types

- Temporary agency worker
- Consultant
- Independent contractor
- Subcontractor employee
- Visitor
- Intern
- Volunteer
- Custom external type

35.2 External worker record

- Person and employing supplier/contractor
- Project/site/role
- Start/end and contract reference
- Induction and competence
- Insurance and access status
- HSE eligibility
- Timesheets where applicable

35.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-EXT-001	External workforce	External workers are not included in payroll unless explicitly configured.	Must	Payroll population filter excludes by default.
HR-EXT-002	External workforce	Supplier/contractor ownership remains visible on every assignment.	Must	No employee misclassification in UI.
HR-EXT-003	External workforce	Eligibility and site access may expire independently from the worker record.	Must	Expired assignment blocks operational use.

36. Asset Integration

HR displays employee-assigned laptops, phones, vehicles, tools, PPE assets, access cards and software licences. Assets remains authoritative for assignment, condition, maintenance, transfer, return and asset history.

ID	Area	Requirement	Priority	Acceptance signal
HR-AST-001	Assets	Employee Page displays authorized asset projections without duplicating asset records.	Must	Asset link opens source page.
HR-AST-002	Assets	Offboarding readiness checks outstanding asset assignments.	Must	Unreturned critical assets are listed as blockers.
HR-AST-003	Assets	Asset return does not automatically close offboarding; HR verifies all requirements.	Must	Source states remain separate.

37. Documents and Record Retention

37.1 Recommended folder structure

```

HR
  /Employees
    /Employee Code - Employee Name
      01 Identity
      02 Employment Contract
      03 Compensation
      04 Attendance
      05 Leave
      06 Payroll
      07 Training
      08 Performance
      09 Disciplinary
      10 Expenses
      11 Onboarding
      12 Offboarding
  /Policies
  /Recruitment
  /Payroll Runs
  /Training
  /Reports

```

37.2 Document controls

- Document Passport
- Revision history
- Required reading
- Expiry
- Confidentiality
- Watermarks
- Retention
- Legal hold
- Read receipts
- Shortcuts without duplicate files

37.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-DOC-001	Documents	HR records link Document IDs and never store independent file copies.	Must	No duplicate blob is created.
HR-DOC-002	Documents	Payslip, contract and certificate visibility follows independent document and HR permissions.	Must	Both checks are enforced.
HR-DOC-003	Retention	Retention actions respect legal hold and employment/case rules.	Must	Held records cannot be destroyed.

38. Offboarding and Final Settlement

38.1 Triggers

- Resignation
- Contract expiry
- Dismissal
- Redundancy
- Retirement
- Mutual agreement
- Death
- Intercompany transfer
- End of assignment

38.2 Checklist

- Final working date and notice
- Leave settlement
- Final timesheet and payroll
- Benefits, loan, advance and expense settlement
- Asset return
- Access revocation
- Document and project responsibility transfer
- Exit interview
- Employment certificate
- Legal documents
- Rehire eligibility

38.3 Data preservation

- Projects
- Tasks
- Documents
- Timesheets
- Payroll
- Attendance
- Decisions
- Team history
- Audit records

38.4 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-OFF-001	Offboarding	Termination and access-revocation effective dates must be coordinated but independently auditable.	Must	Each service records its own state.
HR-OFF-002	Offboarding	Final settlement must show each component and source.	Must	Employee result trace is available.
HR-OFF-003	Offboarding	Active tasks, approvals and owned records require transfer or explicit exception.	Must	Readiness lists unresolved ownership.
HR-OFF-004	Offboarding	Offboarding never deletes historical employment or project contribution.	Must	Historical links remain valid.

39. Intercompany Transfer and Multiple Employment

39.1 Intercompany transfer

40. End the original employment relationship.
41. Preserve original payroll and HR history.
42. Create a new employment relationship.
43. Create/amend contract and compensation.
44. Transfer selected profile information.
45. Re-evaluate leave, benefits, salary and cost centre.
46. Reassign manager, department and project.
47. Preserve User Account where permitted.
48. Re-evaluate access and HSE eligibility.

39.2 Multiple employment

- A Person may hold multiple Employment Relationships.
- Each relationship has separate employer, contract, compensation, payroll group and history.
- Schedules and attendance must detect cross-relationship conflicts where legally/configurably required.
- Payroll remains company-isolated.

ID	Area	Requirement	Priority	Acceptance signal
HR-TRF-001	Transfers	Intercompany transfer never rewrites the original employer.	Must	Old relationship remains closed.
HR-TRF-002	Multiple employment	Payroll calculations are isolated per employment relationship and company.	Must	No cross-company result mixing.
HR-TRF-003	Transfers	Leave transfer treatment requires explicit configured rule and transaction history.	Must	Balances reconcile before/after.

40. Calendar and Meetings Integration

40.1 Calendar projections

- Shifts
- Approved leave
- Interviews
- Training
- Probation and performance reviews
- Contract expiry
- Certificate expiry
- Payroll deadlines
- Onboarding/offboarding events

40.2 Ownership

Calendar owns display and availability projections. HR owns the source dates and lifecycle. Meetings may link to interviews, reviews, disciplinary sessions and return-to-work discussions without owning the HR record.

ID	Area	Requirement	Priority	Acceptance signal
HR-CAL-001	Calendar	Source date changes update projections through idempotent events.	Must	Projection version matches source.
HR-CAL-002	Calendar	Private leave/absence reason is not included in general availability projection.	Must	Only authorized viewers receive detail.

41. Notifications and Workflow Readiness

41.1 Domain events

- EmployeeStarting
- ContractExpiring
- ProbationEnding
- MissingTimesheet
- OvertimeThresholdReached
- LeaveDecisionRequired
- PayrollValidationFailed
- PayrollReady
- PayslipPublished
- CertificateExpiring
- TrainingRequired
- PerformanceReviewDue
- EmployeeLeaving
- AssetReturnOverdue

41.2 Future workflow decision points

- Hiring
- Offer
- Contract
- Salary change
- Leave
- Timesheet
- Payroll
- Bonus
- Expense
- Disciplinary decision
- Termination
- Final settlement

41.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-WFL-001	Events	Events contain tenant, company, actor, correlation and source-version metadata.	Must	Consumer can process idempotently.
HR-WFL-002	Workflow readiness	Records store decision reference without embedding a custom approval engine.	Must	Later integration requires no data migration.
HR-WFL-003	Notifications	HR creates notification intent; future Notifications module owns delivery.	Must	No hard-coded email dependency.

42. Data Model

42.1 Core entities

Person
 Employee
 EmploymentRelationship
 EmploymentStatusHistory
 Position
 PositionAssignment
 Department
 OrganisationUnit
 JobFamily
 Grade
 SalaryBand
 EmploymentContractLink
 CompensationRecord
 CompensationComponent
 BenefitPlan
 EmployeeBenefit
 WorkingSchedule
 ShiftDefinition
 Roster
 ShiftAssignment
 AttendanceEvent
 AttendanceDay
 AttendanceAdjustment
 Timesheet
 TimesheetLine
 LeaveType
 LeaveEntitlement
 LeaveTransaction
 LeaveRequest
 AbsenceCase
 PayrollGroup
 PayrollCalendar
 PayrollPeriod
 PayrollRun
 PayrollEmployeeResult
 PayrollResultLine
 PayrollAdjustment
 Payslip
 PayrollFinanceExport
 EmployeeAdvance
 EmployeeLoan

Garnishment
 ExpenseClaim
 Vacancy
 Candidate
 JobApplication
 Interview
 Assessment
 Offer
 OnboardingPlan
 OnboardingItem
 TrainingCourse
 TrainingRecord
 Qualification
 Certification
 CompetenceRequirement
 PerformanceReview
 PerformanceObjective
 DisciplinaryCase
 GrievanceCase
 WorkforceDemand
 WorkforceAllocation
 ExternalWorker
 OffboardingPlan
 OffboardingItem
 EmployeeDocumentLink
 EmployeeAssetProjection
 HRActivity
 HRMetricSnapshot

42.2 Common effective-date fields

effective_from
 effective_to
 record_version
 created_by
 created_at
 updated_by
 updated_at
 change_reason
 source_reference
 tenant_id
 company_id
 archive_state

42.3 Key constraints

Constraint	Rule
Employment overlap	Configurable; default prevents duplicate active relationship for same person/company/type period.
Primary position	One per employment relationship and effective period.
Base compensation	One active base component per relationship/currency/frequency period.
Payroll result	Unique by run and employment relationship.
Payslip	Unique issued version per payroll result revision.
Leave balance	Derived from immutable leave transactions.
Attendance event	Unique by source/device idempotency key.

43. Backend and Event Architecture

43.1 Recommended services

- Person/Identity Link Service
- Employee Service
- Employment Lifecycle Service
- Organisation and Position Service
- Recruitment Service
- Onboarding Service
- Compensation Service
- Schedule and Roster Service
- Attendance Service
- Timesheet Service
- Leave Service
- Payroll Calculation Engine
- Payroll Validation Service
- Payslip Service
- Payroll Finance Adapter
- Benefits Service
- Training and Competence Service
- Performance Service
- Case Management Service
- Workforce Planning Service
- External Workforce Service
- Offboarding Service
- Document, Asset, HSE, Work Progress, Calendar and Audit adapters

43.2 Event pattern

- Transactional outbox
- Idempotent consumers
- Optimistic concurrency
- Tenant/company scope
- Correlation and causation IDs
- Effective-dated payloads
- Retry and dead-letter handling
- Immutable business events

43.3 Example flow

```
TimesheetPayrollLocked
-> Payroll Input Consumer
-> Validate period and relationship
-> Create versioned time input
-> Recalculate employee result
-> Emit PayrollEmployeeRecalculated
```

43.4 Background jobs

- Payroll calculation and validation
- Leave accrual and expiry
- Certification/contract expiry scans
- Roster conflict scans
- Attendance day build
- Payslip generation
- Finance export/reconciliation
- Retention and legal-hold checks
- Metrics snapshots

44. API Contract

44.1 Employees and organisation

```
GET    /hr/employees
POST   /hr/employees
GET    /hr/employees/{id}
PATCH /hr/employees/{id}
POST   /hr/employees/{id}/transfer
POST   /hr/employees/{id}/terminate
GET    /hr/organisation
POST   /hr/departments
POST   /hr/positions
POST   /hr/positions/{id}/assign
```

44.2 Recruitment and onboarding

```
POST   /hr/vacancies
GET    /hr/vacancies
POST   /hr/candidates
POST   /hr/applications
POST   /hr/offers
POST   /hr/candidates/{id}/convert-to-employee
POST   /hr/onboarding
POST   /hr/offboarding
```

44.3 Attendance, timesheets and leave

```
POST   /hr/attendance/check-in
POST   /hr/attendance/check-out
POST   /hr/attendance/adjustments
GET    /hr/attendance
POST   /hr/timesheets
POST   /hr/timesheets/{id}/submit
POST   /hr/timesheets/{id}/verify
POST   /hr/timesheets/{id}/lock
GET    /hr/leave/balances
POST   /hr/leave/requests
POST   /hr/leave/requests/{id}/cancel
```

44.4 Payroll

```
POST   /hr/payroll/runs
GET    /hr/payroll/runs/{id}
POST   /hr/payroll/runs/{id}/calculate
POST   /hr/payroll/runs/{id}/validate
POST   /hr/payroll/runs/{id}/lock
POST   /hr/payroll/runs/{id}/generate-payslips
POST   /hr/payroll/runs/{id}/export-to-finance
POST   /hr/payroll/runs/{id}/reopen
```

44.5 API standards

- Idempotency key for write actions
- Record version/ETag for updates
- Tenant and company validation
- Permission and field policy context

- Structured validation errors
- Cursor pagination
- Filter/sort schema
- Correlation ID
- Audit metadata
- Rate limiting and request-size controls

45. Frontend Architecture

45.1 Recommended application domains

- HR shell and route registry
- Employee domain
- Organisation graph
- Recruitment workspace
- Roster and attendance workspace
- Timesheet workspace
- Leave workspace
- Payroll workspace
- Training/competence workspace
- Workforce planning workspace
- Self-service workspace

45.2 State management

- Server-state query cache with scoped invalidation
- Form state isolated from source records
- Optimistic updates only for reversible low-risk actions
- No optimistic payroll lock, termination or salary change
- URL-driven filters and saved-view state
- Permission-aware field components
- Version conflict handling and refresh

45.3 Page patterns

Pattern	Use
Module list	Employees, candidates, timesheets, leave, payroll runs and training.
Detail workspace	Employee, candidate, payroll run, vacancy and case pages.
Timeline grid	Rosters, attendance and workforce capacity.
Calculation explorer	Payroll result and component trace.
Wizard	Hire, transfer, payroll run, onboarding and offboarding.
Side drawer	Quick record preview without exposing restricted fields.

46. Security, Privacy and Confidentiality

46.1 Critical access rules

- User Profile access does not grant Employee Record access.
- Manager access does not automatically include salary.
- HR access does not automatically include payroll.
- Payroll access does not automatically include disciplinary or medical records.
- Project access does not expose employment documents.
- Finance receives summaries, not full HR records.
- HSE receives eligibility, not medical detail.

- Employees see only configured self-service fields.
- Sensitive reads are auditable.
- Exports enforce row and field policy.

46.2 Data protection

- Encrypt sensitive fields and documents at rest.
- Mask bank, tax and identification values.
- Separate medical and disciplinary storage/policies.
- Use configurable retention and legal holds.
- Prevent production data in non-production environments.
- Record consent where required.
- Support controlled correction requests.
- Log export and bulk-access events.

46.3 Security requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-SEC-001	Security	Every API enforces tenant, company, row and field policy server-side.	Must	Direct API tests cannot bypass UI.
HR-SEC-002	Security	Sensitive reads and exports are recorded with actor and purpose context where configured.	Must	Audit event is generated.
HR-SEC-003	Security	Payroll and medical data are encrypted with key-rotation support.	Must	Security review passes.
HR-SEC-004	Security	Background jobs execute with scoped service identity, never unrestricted global access.	Must	Job audit includes scope.
HR-SEC-005	Security	Former employee self-service access follows configurable retention and authentication policy.	Must	Access expires or persists explicitly.

47. Employee Self-Service and Manager Workspace

47.1 Employee self-service

- Employment summary
- Contract documents
- Schedule and attendance
- Timesheets
- Leave balance and requests
- Payslips
- Benefits
- Expenses
- Training and certificates
- Onboarding items
- Assigned assets
- Profile-update requests

47.2 Manager workspace

- Team availability
- Upcoming leave
- Attendance exceptions

- Missing timesheets
- Workforce allocation
- Open vacancies
- Probation and review due
- Training gaps
- Contract expirations
- Onboarding/offboarding progress

47.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-ESS-001	Self-service	Employees cannot directly edit authoritative employment or payroll records.	Must	Changes use controlled request flows.
HR-ESS-002	Self-service	Horizontal access to another employee record is denied.	Must	Security tests pass.
HR-MGR-001	Manager	Manager access follows effective reporting relationships and configured scope.	Must	Past/future managers see only allowed periods.
HR-MGR-002	Manager	Salary and confidential-case access remains independent.	Must	Manager page does not infer restricted values.

48. Responsive and Offline UX

48.1 Mobile-critical actions

- Check in/out
- View roster
- Enter/submit timesheet
- Submit leave
- View balance and payslip
- Submit expense and receipt
- View training
- Acknowledge onboarding item
- Confirm asset receipt

48.2 Offline-supported drafts

- Attendance event with device timestamp
- Timesheet draft
- Expense draft
- Receipt/photo
- Training attendance
- Onboarding checklist

48.3 Not finalized offline

- Payroll calculation/lock
- Salary change
- Termination
- Timesheet lock
- Leave decision
- Final settlement
- Intercompany transfer

48.4 Conflict handling

- Client-generated idempotency key
- Device and server timestamps
- Record version check
- Duplicate-event detection
- User-visible sync state
- Conflict review rather than silent overwrite

49. Premium Motion and Accessibility

49.1 Premium interactions

- Employee cards with employment-state transitions
- Organisation chart navigation
- Effective-date timeline
- Roster drag preview with conflict warning
- Attendance anomaly indicators
- Payroll calculation progress
- Expandable calculation trace
- Payslip preview
- Capacity heat maps
- Onboarding/offboarding progress rail
- Skeleton loading

49.2 Accessibility

- WCAG 2.2 AA target
- Keyboard navigation for tables, rosters and dialogs
- Screen-reader labels and live status announcements
- Non-color indicators for payroll errors and employment status
- Reduced-motion mode
- Focus restoration after drawers and modals
- Accessible export alternatives

Privacy UX: Sensitive payroll values must never appear in hover previews, browser notifications, unprotected analytics tooltips or shared screens without explicit reveal.

50. Platform Configuration

Configuration group	Items
Employment	Types, statuses, contract types, organisation levels, numbering.
Compensation	Grades, bands, earnings, deductions, employer costs and currencies.
Time	Schedules, shifts, attendance, GPS/QR policy, overtime and holidays.
Leave	Types, accrual, carry-forward, expiry, attachments and balance rules.
Payroll	Groups, calendars, formulas, taxes, contributions, rounding and validation.
Talent	Recruitment stages, onboarding, training, review and offboarding templates.
Security	Confidentiality, retention, field policies, export and audit settings.

Configuration group	Items
Integration	Notification hooks, approval hooks, Documents, Finance, HSE and Work Progress.

ID	Area	Requirement	Priority	Acceptance signal
HR-CFG-001	Configuration	Every page, section, field, action, rule and report supports Platform Configuration.	Must	Disabled features disappear cleanly.
HR-CFG-002	Configuration	Rule changes are versioned and effective-dated.	Must	Historical payroll retains old version.
HR-CFG-003	Configuration	Configuration promotion between environments is auditable.	Should	Export/import package includes dependencies.

51. Search, Filters and Saved Views

51.1 Search domains

- Employees
- Positions
- Candidates
- Vacancies
- Contracts
- Timesheets
- Leave
- Payroll runs
- Training
- Cases
- Offboarding plans

51.2 Filter capabilities

- Company, department, manager, status, type and location
- Project/work package and cost centre
- Effective-date range
- Contract/certificate expiry
- Attendance and leave state
- Payroll group and validation state
- Confidentiality-aware result filtering

51.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-SRC-001	Search	Search indexing must exclude unauthorized fields and records.	Must	No result-count leakage.
HR-SRC-002	Saved views	Views are private by default and may be shared only with explicit policy.	Must	Sharing does not expand data access.
HR-SRC-003	Exports	Export uses the same row and field security as the current query.	Must	Export cannot reveal hidden columns.

52. Reporting and Analytics

52.1 HR reports

- Headcount and employee register
- New hires and turnover
- Contract expiry
- Vacancies and recruitment funnel
- Attendance and absence
- Leave balances
- Overtime
- Timesheet completion
- Training and certification
- Workforce allocation
- Employee and project labour cost
- Benefits
- Performance-review completion
- Offboarding

52.2 Payroll reports

- Gross-to-net
- Payroll register
- Variance
- Earnings and deductions
- Employer costs
- Taxes and contributions
- Project/department allocation
- Payment reconciliation
- Final settlement
- Correction history

52.3 Analytics rules

- Every metric drills into source records.
- Currency and company context are explicit.
- Estimated values are distinguished from locked payroll and posted Finance values.
- Operational metrics are not presented as subjective performance judgments.
- Sensitive reports require independent permissions.

53. Validation, Errors and Concurrency

53.1 Validation layers

49. Schema and type validation.
50. Tenant/company and permission validation.
51. Effective-date and lifecycle validation.
52. Cross-domain business validation.
53. Payroll rule validation.
54. Concurrency/version validation.
55. External-adaptor response validation.

53.2 Error contract

```
{
  "code": "PAYROLL_LOCK_BLOCKED",
  "message": "Payroll run has unresolved validation errors.",
  "details": [{"recordId": "...", "rule": "NET_PAY_NEGATIVE"}],
  "correlationId": "...",
}
```

```
"retryable": false
}
```

53.3 Concurrency requirements

- Optimistic version checks on authoritative records.
- Serializable or protected transactions for payroll lock, leave ledger and unique assignments.
- Idempotency for check-in, imports, payroll calculation and Finance export.
- No silent last-write-wins on salary, employment, payroll or termination.

54. Non-Functional Requirements

Category	Requirement
Availability	Core HR read operations target 99.9% monthly availability excluding planned maintenance.
Performance	P95 employee list under 2 seconds for normal scoped datasets; detail under 2.5 seconds.
Payroll scale	Support at least 25,000 employees per payroll group with resumable batch calculation.
Attendance scale	Sustain high-volume check-in bursts with idempotent ingestion.
Data integrity	ACID transactions for payroll lock, leave ledger, employment transitions and Finance export state.
Security	Encryption in transit/at rest, least privilege, audit and secrets management.
Accessibility	WCAG 2.2 AA target.
Localization	Multiple languages, currencies, time zones, holiday calendars and number/date formats.
Recovery	Documented backup, point-in-time recovery and disaster-recovery tests.
Retention	Configurable retention and legal hold without orphaning linked records.

54.1 Suggested performance budgets

- Dashboard aggregate P95 <= 3.0 seconds.
- Payroll employee result detail P95 <= 2.5 seconds.
- Attendance ingestion acknowledgement P95 <= 500 ms under normal load.
- Search P95 <= 2.0 seconds for permitted result sets.
- Bulk import validates asynchronously and reports row-level errors.

55. Observability and Audit

55.1 Audit events

- Employee creation/update
- Employment transition
- Position assignment
- Compensation change
- Attendance correction
- Timesheet status
- Leave transaction

- Payroll calculation/validation/lock/reopen
- Payslip issue
- Finance export
- Sensitive read/export
- Case access
- Offboarding and transfer

55.2 Technical observability

- Structured logs with correlation IDs
- Metrics by service, company and job type
- Distributed traces for payroll and cross-module flows
- Job progress and retry visibility
- Dead-letter monitoring
- Security anomaly alerts
- Data-reconciliation dashboards

55.3 Requirements

ID	Area	Requirement	Priority	Acceptance signal
HR-AUD-001	Audit	Business audit events are immutable and separate from application logs.	Must	Users cannot edit/delete audit history.
HR-AUD-002	Audit	Sensitive read/export events record actor, scope and source.	Must	Audit query can reconstruct access.
HR-OBS-001	Observability	Payroll jobs expose progress, failure phase and retry state.	Must	Operations can resume safely.

56. QA Strategy and Test Matrix

56.1 Required test suites

- Unit and domain tests
- API contract tests
- Employment lifecycle tests
- Effective-date tests
- Schedule/DST/cross-midnight tests
- Attendance duplicate/offline tests
- Leave accrual ledger tests
- Payroll gross-to-net, rounding and retro tests
- Finance reconciliation tests
- Multi-company and multi-currency tests
- Privacy and horizontal-access tests
- Offboarding preservation tests
- Mobile/responsive tests
- Accessibility tests
- Performance and high-volume tests
- Backup/recovery tests

56.2 Critical scenarios

Scenario	Expected behavior
Salary changes mid-period	Result prorates or applies configured rule with full trace.
Employee transfers between companies	Original relationship closes; new company history starts.
Employee works on multiple projects	Hours and cost allocate without changing employer.

Scenario	Expected behavior
Leave overlaps night shift	Configured schedule/date policy produces deterministic hours.
Timesheet changes after payroll lock	Adjustment flow required; original result remains.
Termination backdated	Impact analysis and controlled payroll/access correction.
Payroll posted then reopened	Finance reversal/adjustment linkage required.
Expired certificate	Roster or HSE assignment blocked according to policy.
Manager views attendance	Salary remains hidden.
Offboarded employee owns active tasks	Transfer or explicit exception required.
Offline check-in retried	Single attendance event is created.
Bank details change during payment prep	Configured freeze/version check prevents silent destination change.

57. Acceptance Criteria

ID	Acceptance criterion
AC-001	A person can be linked to multiple historical employment relationships without data loss.
AC-002	Employee and User Profile remain separate permission domains.
AC-003	Every employment, position, schedule and compensation change is effective-dated.
AC-004	Historical date queries reproduce the state effective on that date.
AC-005	Duplicate active employment and primary-position overlaps are prevented.
AC-006	Candidate conversion preserves recruitment history and avoids duplicate files.
AC-007	Onboarding blockers are visible and linked to authoritative source modules.
AC-008	Signed contract versions are immutable and amendments preserve history.
AC-009	Attendance raw events are immutable and corrections are auditable.
AC-010	Offline attendance retries do not create duplicates.
AC-011	Roster conflicts are detected using schedule, leave, eligibility and employment dates.
AC-012	Timesheet hours are project/work-package allocatable and payroll-lockable.
AC-013	Locked timesheets are corrected only through adjustment records.
AC-014	Leave balances rebuild from an immutable transaction ledger.
AC-015	Leave availability projections do not expose private reasons.
AC-016	Medical details are separated from operational fitness status.
AC-017	Payroll calculation is deterministic for versioned inputs.
AC-018	Every payroll line exposes rule, source, quantity, rate, formula and rounding.
AC-019	Payroll hard errors block lock and warnings require recorded resolution.

ID	Acceptance criterion
AC-020	Locked payroll results cannot be directly edited.
AC-021	Retroactive changes create linked adjustments rather than silent rewrites.
AC-022	Payslips can be generated only from locked payroll results.
AC-023	Employees can access only their own payslips through self-service.
AC-024	Payroll-to-Finance export is balanced, idempotent and reconcilable.
AC-025	Finance receives no unrelated confidential HR data.
AC-026	Loans, advances and garnishments reconcile to immutable balance transactions.
AC-027	Training and competence expiry affects eligibility according to configuration.
AC-028	HSE eligibility APIs expose only necessary operational data.
AC-029	Task, attendance and activity data do not automatically create performance ratings.
AC-030	Disciplinary and grievance records have independent access controls.
AC-031	Workforce capacity calculations expose their source values.
AC-032	External workers are excluded from payroll unless explicitly configured.
AC-033	Offboarding checks assets, access, payroll, responsibilities and documents.
AC-034	Offboarding preserves all historical work and audit links.
AC-035	Intercompany transfer creates a new employment relationship.
AC-036	Calendar projections remain source-linked and privacy-aware.
AC-037	Search and exports enforce row and field permissions.
AC-038	Sensitive reads and exports are auditable.
AC-039	All write APIs support idempotency and optimistic versioning where applicable.
AC-040	Platform Configuration can disable features without broken navigation or API leakage.
AC-041	Mobile attendance and timesheet drafts synchronize with visible conflict handling.
AC-042	The module supports multiple companies, currencies, time zones and holiday calendars.
AC-043	Payroll scale and performance tests meet the stated budgets.
AC-044	Legal hold prevents destruction of protected HR records.
AC-045	Audit and reconciliation reports can reconstruct payroll from source to Finance posting.

58. Delivery Phases and Migration

58.1 Phase 1 - HR foundation

- Employee records

- Organisation and positions
- Employment history
- Contracts
- Recruitment
- Onboarding
- HR documents
- Self-service
- Audit

58.2 Phase 2 - Workforce operations

- Schedules
- Shifts/rosters
- Attendance
- Timesheets
- Leave
- Project labour
- Training/competence
- Assets/HSE integration

58.3 Phase 3 - Payroll

- Compensation
- Payroll groups/calendars/rules
- Payroll runs
- Payslips
- Advances/deductions
- Final settlement
- Finance integration
- Payroll reporting

58.4 Phase 4 - Talent and advanced workforce

- Performance
- Disciplinary/grievances
- Benefits
- Workforce forecasting
- External workforce
- Advanced analytics
- Offline operations

58.5 Migration approach

56. Define company, organisation, position and payroll configuration.
57. Import people and employees with deterministic external IDs.
58. Import effective-dated employment, contract and compensation history.
59. Reconcile opening leave balances through opening ledger transactions.
60. Import current schedules, assignments, certificates and assets.
61. Import payroll history as locked historical snapshots where detailed source data is unavailable.
62. Run parallel payroll and reconciliation before production cutover.
63. Freeze legacy changes, execute delta migration and validate totals.

59. Risks and Open Decisions

Risk / decision	Required resolution
Payroll jurisdiction rules	Define per-country configuration ownership, legal review and test datasets before payroll release.
Cost-rate privacy	Decide standard-cost vs actual-cost projection policy for Projects and Work Progress.

Risk / decision	Required resolution
Multiple employment	Confirm permitted overlap and schedule conflict policy by legal company/jurisdiction.
GPS attendance	Define lawful purpose, consent/notice, retention and geofence precision by site.
Retro payroll	Confirm reopening vs current-period adjustment policy and Finance reversal flow.
Leave transfer	Define group-company transfer rules for entitlement, carry-over and payout.
External workers	Define which worker types require timesheets, access and competence records.
Medical records	Define minimum operational fitness model and retention/access boundaries.
Former employee access	Decide payslip/document access after termination.
Approvals dependency	Map decision hooks but defer orchestration until the final Workflow PRD.

60. Page and Flow Registry

60.1 Primary pages

Code	Page	Primary responsibility
HR-001	HR Dashboard	Cross-domain workforce overview and attention centre.
HR-002	Employees Module	Search, filter, bulk and saved views.
HR-003	Employee Page	Complete employment workspace.
HR-004	Organisation Page	Hierarchy, positions and headcount.
HR-005	Vacancy Page	Hiring requirement and candidate pipeline.
HR-006	Candidate Page	Candidate history, assessments and offers.
HR-007	Onboarding Page	Readiness, checklist and integrations.
HR-008	Roster Page	Shift planning, conflicts and capacity.
HR-009	Attendance Page	Raw events, daily states and anomalies.
HR-010	Timesheet Page	Time allocation and payroll lock.
HR-011	Leave Page	Requests, balances and transactions.
HR-012	Payroll Dashboard	Run calendar, controls and validation.
HR-013	Payroll Run Page	Calculation, trace, reconciliation and export.
HR-014	Training & Competence	Requirements, gaps and eligibility.
HR-015	Workforce Planning	Demand, supply, allocation and cost.
HR-016	Case Page	Disciplinary or grievance restricted workspace.
HR-017	Offboarding Page	Readiness, settlement and transfer.
HR-018	Self-Service	Employee-owned permitted views and requests.
HR-019	Manager Workspace	Team operational HR controls.

Code	Page	Primary responsibility
HR-020	HR Settings	Configuration, templates and rule versions.

60.2 Critical flows

Flow	Start	End
Hire	Accepted candidate	Active employee with onboarding plan.
Position change	Change request/context	Effective-dated assignment and compensation impact.
Attendance day	Raw check events	Derived daily attendance state.
Timesheet to payroll	Verified time	Payroll-locked input.
Leave	Employee request	Balance transaction and calendar projection.
Payroll	Open period	Closed period with Finance/payment references.
Retro correction	Effective-dated change	Linked payroll adjustment.
Intercompany transfer	Active employment	Closed old + active new relationship.
Offboarding	Termination trigger	Former employee with settled responsibilities.

Implementation conclusion: This module must be implemented as an employment and payroll operating system, not an employee directory. The permanent architecture is: User Profile represents the person professionally; HR governs employment; Payroll calculates compensation; Work Progress records labour usage; HSE governs eligibility; Finance records accounting and payment.